C++ Set-Up

Intro to Audio Tech 2
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Lesson Outline

- Setting up our environment for C++
- C++ overview
- Up Next…

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C++ Set-Up

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IDE and JUCE Install

- Mac
  - Install Xcode from the app store
- Windows
  - Install Visual Studio 2026 (not the same thing as VS code) from https://visualstudio.microsoft.com
- Everyone
  - Install JUCE from https://juce.com

C++ Overview

Why C++?

- C++ is the language most used for audio programming and development
- It allows you to interface directly with the computer memory and soundcards
- C++ is an extension on C with better features for developers (object oriented)
  - Both low-level control along with high-level programming structure
- It is used for audio programming as well as video game development and general system development

Why C++?

- When building audio applications, speed is crucial (especially for real-time systems)
- C++ is “close to hardware” which lets us write efficient code directly to the machine
- The high level characteristics still let us write modular, “block-based” code

Differences from Python

- Python is an interpreted language while C++ is compiled
- Compiled
  - Our semantic code is translated into machine code that will run directly on the CPU
- Interpreted
  - Our semantic code is run line by line and “interpreted” from text to machine code by an additional program (the python interpreter)
- In C++, we have to allocate memory and declare variables, arrays, and functions differently than in Python
- The syntax in general is a bit different, but concepts carry over

Syntax examples

	Python	C++
Variable declaration	myVar = 1 myString = ”hello”	int myVar = 1; std::string myString = ”hello”;
Arrays	myArray = np.array([0, 2, 3, 4])	int myArray[4] myArray = {0, 2, 3, 4}; -or- std::array<int, 4> myArray = {0, 2, 3, 4};

Functions	myFunc(num1, num2):
	  myVal = num1 + num2
	  return myVal
		int myFunc(int num1, int num2) {
		  int myVal = num1 + num2;
		  return myVal;

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C++ Projects

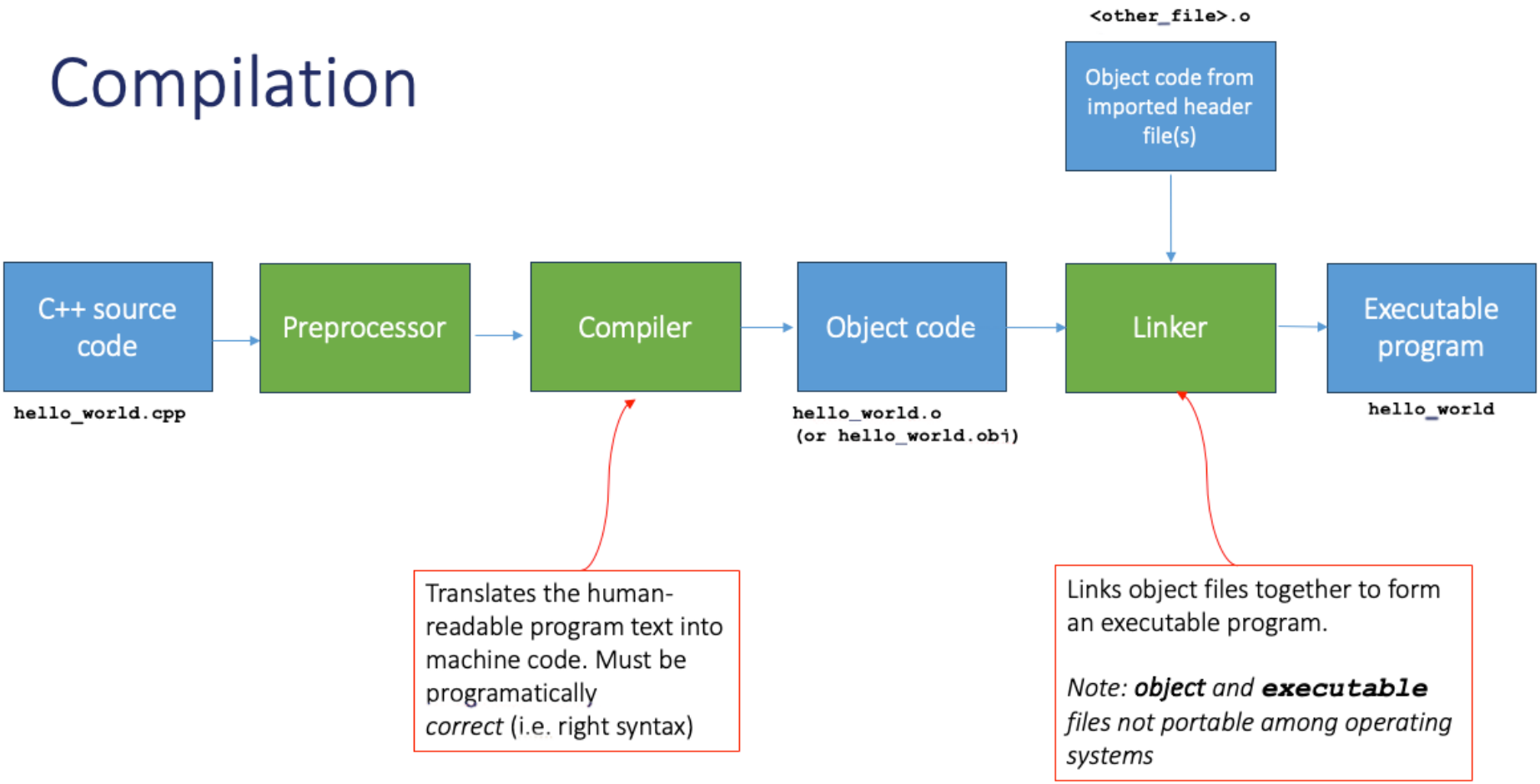
- Each project we work on will have a collection of files
  - Source Files (.cpp) – this is where we write the majority of our code
  - Header Files (.h) – this is where additional definitions/functions go
  - A project file (.xcodeproj or .vcxproj) – this is for our specific IDE and has our build settings
  - A jucer file (.juce) – this is for additional build settings and juce

Build Process

- Because our code has to be compiled into machine code, there are a few steps to the process
- The build phase is typically the more annoying part of writing C/C++
  - the majority of the annoying parts should be handled by JUCE

1. Pre-Processing: All header files and macros are expanded into the source files
2. Compilation: The compiler ”transforms” source code to assembly code (still human readable but low-level)
3. Assembly: The assembly code is turned into machine-readable binaries
4. Linking: The linker fills in the binaries with any missing library code and puts everything together into an executable

Compilation



JUCE

JUCE

- JUCE is an open-source framework that helps build software across many operating systems
- JUCE is helpful because it deals with all the build process specifications for us and includes a lot of libraries that help that build process go smoothly
- Also, JUCE helps us keep the same code across any final build target
  - We can use the same code for a VST or an AU and JUCE will handle the build process

JUCE

Sine Wave Generator and Gain Plugin

- Let’s get started with our first project in JUCE
- To test all of our systems, we are just going to look at a ProJucer example project today
- Some notes for revisiting this later:
  - Open the ProJucer
  - File -> New Project
  - Open Example -> AudioAppDemo
  - Go to modules and make sure ProJucer knows the path to your JUCE modules
  - Go to exporters and make sure you are building to at least your IDE
  - Select the correct exporter at the top and hit the button to open the project in your IDE
  - Build and run the project in your IDE